RGF signaling coordinates redox homeostasis and root meristem development under non-lethal heat stress

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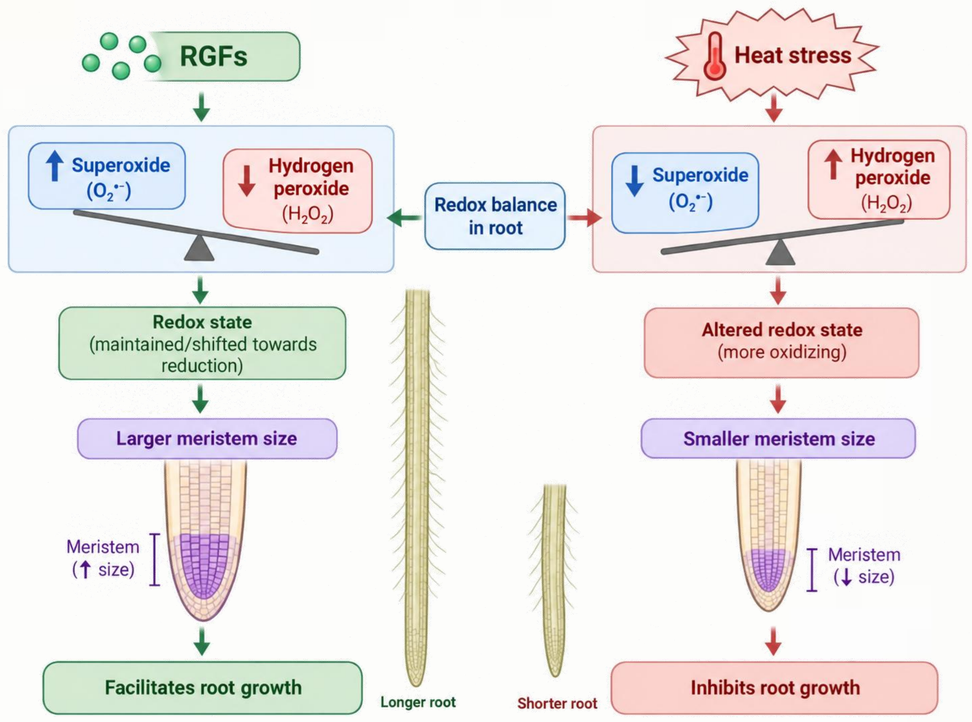
Background:

Elevated temperatures increasingly threaten plant growth, yet how roots respond to non-lethal heat stress (NLH) remains poorly understood. In *Arabidopsis thaliana*, root growth depends on the control of root meristem size, which reflects the balance between cell proliferation and differentiation. Meristem size is regulated by the RGF—RGFR—PLT2 signaling pathway and by the redox balance between superoxide ($O_2^{\bullet-}$) and hydrogen peroxide (H_2O_2). While severe heat stress activates canonical heat-shock responses, how NLH affects meristem size control remains unclear.

Question: How does NLH regulate root meristem size through RGF signaling and redox homeostasis?

Objectives:

- Define the effects of NLH on primary root growth, meristem size, and cellular redox status.
- Determine whether NLH alters RGF—RGFR—PLT2 signaling and whether this pathway is required under heat stress.
- Test whether exogenous RGF peptides mitigate NLH-associated primary and lateral root defects.



Summary:

- **Non-lethal heat stress (NLH) suppresses root growth without causing plant death.** Exposure to 31°C reduced primary root growth, meristem size, and superoxide ($O_2^{\bullet-}$) levels. Imaging with a redox biosensor revealed a shift of the root meristem toward a more oxidized state. These changes were associated with premature differentiation and reduced cell proliferation.
- **NLH represses the RGF—RGFR—PLT2 signaling pathway and disrupts redox homeostasis.** Transcriptome analysis revealed preferential down-regulation of meristem-associated genes and upregulation of differentiation-related genes rather than activation of canonical heat-shock responses.
- **RGF—RGFR—PLT2 signaling is required for root adaptation to NLH.** Mutants defective in RGF, RGFR, or PLT2 signaling exhibited increased sensitivity to NLH, indicating an important role for this pathway in sustaining root meristem size under thermal stress.
- **Exogenous RGF peptides alleviate NLH-induced developmental defects.** RGF1 treatment restored root meristem defects and redox-associated changes, while RGF2 and RGF8 promoted lateral root initiation and elongation under NLH.

Non-lethal heat stress (NLH) restricts root growth and alters redox homeostasis

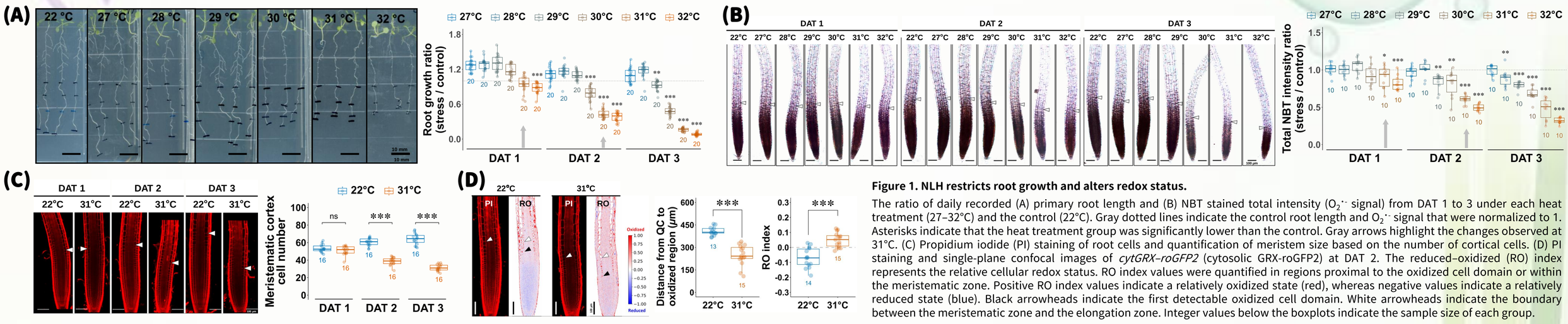


Figure 1. NLH restricts root growth and alters redox status.

The ratio of daily recorded (A) primary root length and (B) NBT stained total intensity ($O_2^{\bullet-}$ signal) from DAT 1 to 3 under each heat treatment (27–32°C) and the control (22°C). Gray dotted lines indicate the control root length and $O_2^{\bullet-}$ signal that were normalized to 1. Asterisks indicate that the heat treatment group was significantly lower than the control. Gray arrows highlight the changes observed at 31°C. (C) Propidium iodide (PI) staining of root cells and quantification of meristem size based on the number of cortical cells. (D) PI staining and single-plane confocal images of *cytGRX-roGFP2* (cytosolic GRX-roGFP2) at DAT 2. The reduced–oxidized (RO) index represents the relative cellular redox status. RO index values were quantified in regions proximal to the oxidized cell domain or within the meristematic zone. Positive RO index values indicate a relatively oxidized state (red), whereas negative values indicate a relatively reduced state (blue). Black arrowheads indicate the first detectable oxidized cell domain. White arrowheads indicate the boundary between the meristematic zone and the elongation zone. Integer values below the boxplots indicate the sample size of each group.

NLH up-regulates differentiation-related genes and down-regulates meristem-associated genes

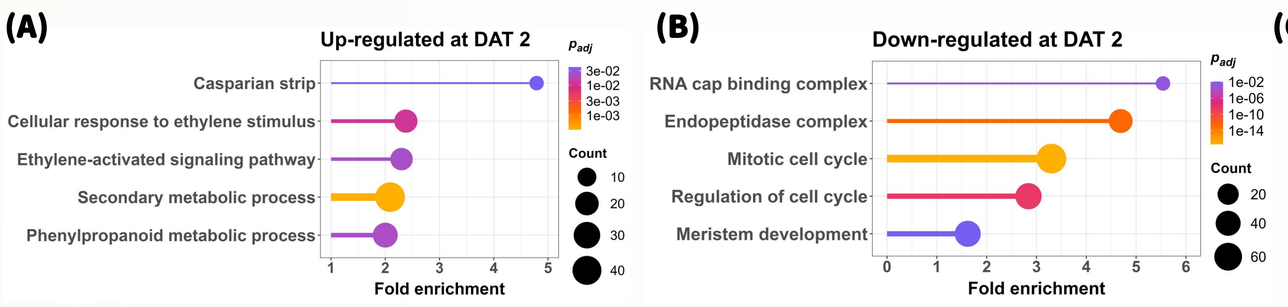


Figure 2. NLH promotes differentiation-related genes and suppresses meristem-associated genes.

Gene Ontology (GO) analysis of (A) up-regulated and (B) down-regulated genes at DAT 2. The GO terms are biological processes that were filtered by fold-enrichment, p-adjusted value, and total gene counts. (C) RNA-seq-based quantification of the gene expression of RGF–RGFR–PLT2 signaling pathway-related genes after treated with 22°C or 31 °C for 1 or 2 days. RGF2: root meristem growth factor 2, RGF8: root meristem growth factor 8, PLT2: PLETHORA 2, CYCB1;3: CYCLIN B1;3, RGFR2: RGF1 INSENSITIVE 2, TPST: TYROSYLPROTEIN SULFOTRANSFERASE. DAT, days after treatment.

RGF-pathway mutants are hypersensitive to NLH

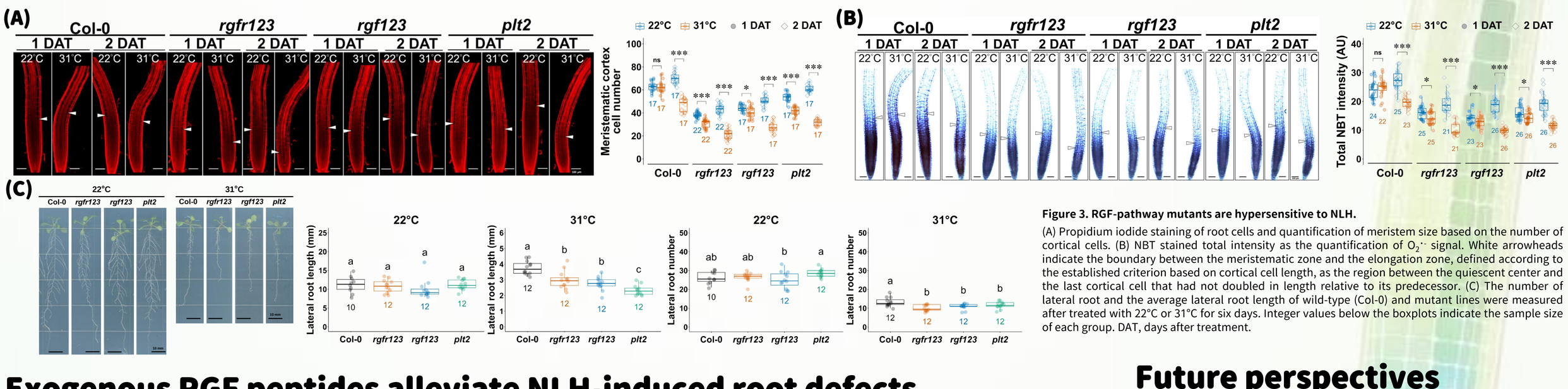


Figure 3. RGF-pathway mutants are hypersensitive to NLH.

(A) Propidium iodide staining of root cells and quantification of meristem size based on the number of cortical cells. (B) NBT stained total intensity as the quantification of $O_2^{\bullet-}$ signal. White arrowheads indicate the boundary between the meristematic zone and the elongation zone, defined according to the established criterion based on cortical cell length, as the region between the quiescent center and the last cortical cell that had not doubled in length relative to its predecessor. (C) The number of lateral root and the average lateral root length of wild-type (Col-0) and mutant lines were measured after treated with 22°C or 31°C for six days. Integer values below the boxplots indicate the sample size of each group. DAT, days after treatment.

Exogenous RGF peptides alleviate NLH-induced root defects

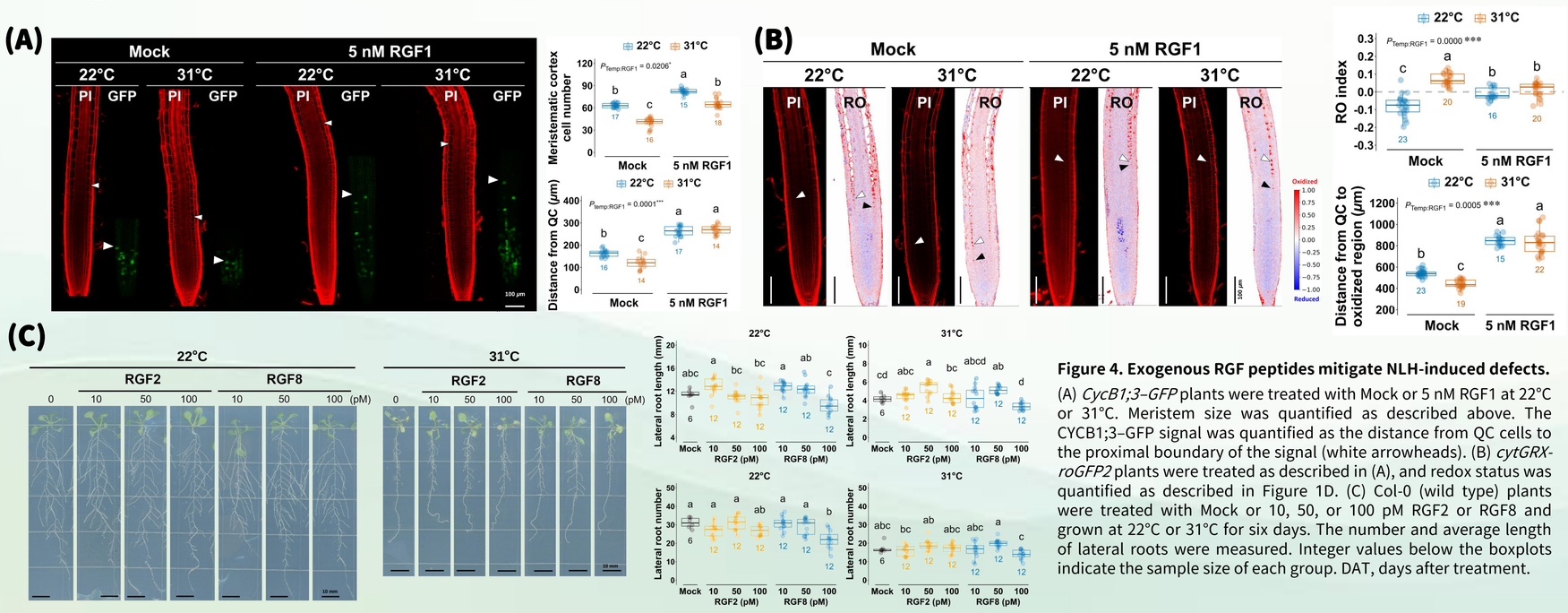


Figure 4. Exogenous RGF peptides mitigate NLH-induced defects.

(A) *CycB1;3–GFP* plants were treated with Mock or 5 nM RGF1 at 22°C or 31°C. Meristem size was quantified as described above. The CYCB1;3–GFP signal was quantified as the distance from QC cells to the proximal boundary of the signal (white arrowheads). (B) *cytGRX-roGFP2* plants were treated as described in (A), and redox status was quantified as described in Figure 1D. (C) Col-0 (wild type) plants were treated with Mock or 10, 50, or 100 pM RGF2 or RGF8 and grown at 22°C or 31°C for six days. The number and average length of lateral roots were measured. Integer values below the boxplots indicate the sample size of each group. DAT, days after treatment.

Future perspectives

This study identifies RGF—RGFR—PLT2 signaling as a key developmental module that links redox regulation with root meristem maintenance under non-lethal heat stress. Future work will investigate how heat signals are sensed and transmitted to RGF peptides, as well as how redox changes modulate RGFR activity and PLT2 stability or spatial distribution. Identifying downstream redox regulators—such as thioredoxins, glutaredoxins, and reactive oxygen species–producing enzymes—will further clarify the molecular framework connecting oxidative status to meristem activity. In addition, evaluating RGF treatment in crop species and under fluctuating field temperatures will help assess its translational potential for improving root thermotolerance and yield stability. Finally, engineering or selecting genotypes with enhanced RGF signaling may provide an effective strategy for developing climate-resilient crops.

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